

Drawing DAGs: puerperal fever

Drawing DAGs

Allow me to repeat myself: DAGs can be a wonderfully useful tool for thinking about causation. We're going to practice with the puerperal fever example we have just read about. I would like you to, *at the whiteboard*, draw the DAG for Ignaz Semmelweis' understanding of puerperal fever in the maternity ward¹.

Here are a list of the variables (nodes) of interest. Please use them to draw a DAG

- Bleach water
- Cadaver particles on hands
- Doctor's vs. midwives ward
- Puerperal fever
- Childbirth
- Doctor laziness
- Autopsies conducted

Here are some hints and reminders:

- The variables or “nodes” represent things that you could potentially measure or states of the system (e.g., “temperature” or “disease”)
 - they are not processes (i.e., not “infection”, but “infection status”)
 - amounts of something (i.e., not “increased temperature,” but “temperature”)

¹ Not how we, in modern times, would view it.

- The arrows represent causation, such that $x \rightarrow y$ reads as, “a change in x *causes* a change in y ”
 - the arrows do not mean “increase” or “decrease” (i.e., x could be increased or decreased and y would respond accordingly)
- It can be helpful to work backwards with the effect (e.g., if “infection status” is the effect of interest, then ask, what things control whether an individual is infected or not?)